

E1 Trigonometry — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

E1_Q1: Pythagoras Theorem

Hook

Have you ever needed to find the shortest distance between two points when you can't walk in a straight line? Architects use Pythagoras' Theorem every day to calculate diagonal roof lengths, TV screen sizes are quoted as the diagonal measurement (which uses Pythagoras), and GPS systems use it to calculate distances between coordinates. This simple but powerful relationship between the sides of a right-angled triangle has been used for thousands of years.

Prerequisites

- Identifying right-angled triangles (90-degree angle)
- Squaring numbers and finding square roots
- Understanding that the hypotenuse is the longest side opposite the right angle
- Basic algebraic rearrangement of formulas

Core Explanation

What is Pythagoras' Theorem?

Pythagoras' Theorem describes the relationship between the three sides of a **right-angled triangle**. It states that:

In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$a^2 + b^2 = c^2$$

Where:

- **c** = the **hypotenuse** (the longest side, opposite the right angle)
- **a** and **b** = the two shorter sides (legs)

Important Notes:

- The theorem **only** works for right-angled triangles
- The hypotenuse is always opposite the 90-degree angle and is the **longest** side
- You can use the theorem to find any missing side if you know the other two

Finding the Hypotenuse (c):

$$c = \sqrt{a^2 + b^2}$$

Finding a Shorter Side (a or b):

$$a = \sqrt{c^2 - b^2}$$

$$b = \sqrt{c^2 - a^2}$$

3-4-5 Triangle Rule

A well-known Pythagorean triple is 3-4-5: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$

This means any triangle with sides 3, 4, 5 (or multiples like 6, 8, 10) is right-angled. Other common triples: 5-12-13, 8-15-17, 7-24-25.

Worked Example

Problem: A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm. Find the length of the other shorter side.

Step 1: Identify what we know.

- Hypotenuse (c) = 13 cm
- One shorter side (a) = 5 cm
- Unknown side (b) = ?

Step 2: Write down Pythagoras' Theorem.

$$a^2 + b^2 = c^2$$

Step 3: Substitute the known values.

$$5^2 + b^2 = 13^2$$

Step 4: Calculate the squares.

$$25 + b^2 = 169$$

Step 5: Rearrange to isolate b^2 .

$$b^2 = 169 - 25$$

$$b^2 = 144$$

Step 6: Take the square root (positive value, as length is positive).

$$b = \sqrt{144}$$

$$b = 12 \text{ cm}$$

Answer: The other shorter side is 12 cm.

Check: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$. Correct!

Common Mistakes

X Applying Pythagoras' Theorem to non-right-angled triangles

OK Fix: Pythagoras only works for triangles with a 90-degree angle. For non-right triangles, you must use the Sine Rule.

X Using the wrong side as the hypotenuse in the formula (e.g., $a^2 + c^2 = b^2$)

OK Fix: The hypotenuse must always be 'c' in the formula $a^2 + b^2 = c^2$. Identify the longest side first and label it 'c'.

X Forgetting to take the square root at the end, leaving the answer as c^2 instead of c

OK Fix: Pythagoras gives $c^2 = a^2 + b^2$. You must take the square root to find c . Always check: is my answer

Summary

- Pythagoras' Theorem ($a^2 + b^2 = c^2$) relates the sides of a right-angled triangle, where c is the hypotenuse (longest side).
- To find the hypotenuse: $c = \sqrt{a^2 + b^2}$. To find a shorter side: $a = \sqrt{c^2 - b^2}$.
- The theorem only applies to right-angled triangles — check for the 90-degree angle before using it.
- Common Pythagorean triples (e.g., 3-4-5, 5-12-13) can help you spot right-angled triangles quickly.
- Always include units in your final answer and check by substituting back into the formula.

E1_Q2: Trig Ratios SOH CAH TOA

Hook

How does a pilot know how far to descend before landing, or how does a surveyor measure the height of a mountain without climbing it? The answer lies in the three trigonometric ratios: sine, cosine, and tangent. These ratios unlock the relationship between angles and side lengths in right-angled triangles, allowing us to measure the unmeasurable. The mnemonic SOH CAH TOA is your key to remembering them.

Prerequisites

- Identifying the hypotenuse, opposite, and adjacent sides in a right-angled triangle relative to a given angle
- Basic understanding of angles and how they are measured in degrees
- Using a scientific calculator to find $\sin$, $\cos$, $\tan$ of an angle and their inverses ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$)
- Rounding decimal answers to a specified number of decimal places or significant figures

Core Explanation

Trigonometric Ratios

In any right-angled triangle, the ratios between side lengths depend only on the size of the acute angles, not on the size of the triangle itself. These ratios are called **sine**, **cosine**, and **tangent**.

Labelling the Triangle

For a given angle θ (written as ' θ '):

- **H** = **Hypotenuse** — longest side, opposite the right angle (always the same)
- **O** = **Opposite** — the side directly across from angle θ
- **A** = **Adjacent** — the side next to angle θ (but not the hypotenuse)

SOH CAH TOA

Mnemonic	Ratio	Formula
----- ----- -----		
S	$\sin(\theta) = O / H$	$\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$
C	$\cos(\theta) = A / H$	$\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$
T	$\tan(\theta) = O / A$	$\tan(\theta) = \text{Opposite} / \text{Adjacent}$

Using the Ratios

There are two types of problems:

Type 1: Finding a side length — you know an angle and one side, need another side.

- Identify which sides are involved (O, A, or H)
- Choose the correct ratio (SOH, CAH, or TOA)
- Substitute and solve

Type 2: Finding an angle — you know two sides, need an angle.

- Identify which sides you know relative to the unknown angle
- Choose the correct ratio
- Use the inverse trigonometric function ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) on your calculator

Exact Trigonometric Values

You should memorise these for your IGCSE exam:

Angle	sin	cos	tan
----- ----- -----			
0 degrees	0	1	0
30 degrees	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45 degrees	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60 degrees	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90 degrees	1	0	undefined

Worked Example

Problem: In a right-angled triangle, the angle is 30 degrees and the hypotenuse is 10 cm. Find the length of the side opposite the 30-degree angle.

Step 1: Label the triangle relative to the 30-degree angle.

- Opposite = ? (what we want to find)
- Hypotenuse = 10 cm (given)
- The sides involved are Opposite and Hypotenuse

Step 2: Choose the correct ratio.

- SOH uses O and H: $\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$

****Step 3:**** Substitute the known values.

$$- \sin(30 \text{ degrees}) = \text{Opposite} / 10$$

****Step 4:**** Use known value or calculator.

$$- \sin(30 \text{ degrees}) = 0.5$$

$$- 0.5 = \text{Opposite} / 10$$

****Step 5:**** Solve for Opposite.

$$- \text{Opposite} = 0.5 \times 10$$

$$- \text{Opposite} = 5 \text{ cm}$$

****Answer:**** The side opposite the 30-degree angle is 5 cm.

****Check:**** Using exact values, $\sin(30) = 1/2$, so $1/2 = O/10$, thus $O = 5$. Correct!

Common Mistakes

X Confusing which side is opposite vs adjacent when the angle changes

OK Fix: The opposite and adjacent sides RELATIVE to the chosen angle. Always label your triangle for the specific angle.

X Using sin, cos, or tan when the calculator mode is in radians instead of degrees

OK Fix: Always check your calculator is set to DEGREES mode (usually shown as 'D' or 'DEG' on screen). In radians mode, you would get a different result.

X Forgetting to use inverse trig functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) when finding an angle

OK Fix: To find an angle from a ratio, use the inverse function. For example, if $\sin(\theta) = 0.5$, then $\theta = \sin^{-1}(0.5)$.

Summary

- SOH CAH TOA is the mnemonic: $\sin = O/H$, $\cos = A/H$, $\tan = O/A$ for a right-angled triangle.
- Always label the triangle sides (O, A, H) relative to the angle you are working with before choosing a ratio.
- To find a missing side: pick the correct ratio, substitute known values, and solve algebraically.
- To find a missing angle: use the inverse trig functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) on your calculator in degree mode.
- Memorise the exact values for sin, cos, and tan of 0, 30, 45, 60, and 90 degrees for exam efficiency.

E1_Q3: Angles of Elevation/Depression

Hook

A park ranger spots a fire on a distant mountain. How far away is it? A sailor on a lighthouse sees a boat approaching — how far is the boat from the base of the lighthouse? These real-world problems involve measuring angles of elevation and depression. By combining these angles with trigonometric ratios, we can calculate distances and heights without ever leaving our vantage point. This is the mathematics

behind surveying, navigation, and search-and-rescue operations.

Prerequisites

- Understanding of SOH CAH TOA trigonometric ratios
- Ability to identify right-angled triangles in practical contexts
- Using a calculator to find trigonometric values and inverse trig functions
- Interpreting diagrams drawn from word problems

Core Explanation

What Are Angles of Elevation and Depression?

Angle of Elevation

The **angle of elevation** is the angle measured **upward** from the horizontal line of sight to an object above the observer.

- When you look up at a plane, the angle between the horizontal and your line of sight is the angle of elevation
- Measured from the horizontal **upwards**

Angle of Depression

The **angle of depression** is the angle measured **downward** from the horizontal line of sight to an object below the observer.

- When you look down from a cliff to a boat, the angle between the horizontal and your line of sight is the angle of depression
- Measured from the horizontal **downwards**

Key Relationship: Alternate Angles

A crucial insight: **The angle of depression from point A to point B is equal to the angle of elevation from point B to point A** (alternate angles on parallel lines are equal).

Setting Up Problems

1. Draw a diagram showing the observer, the object, and the horizontal line
2. Mark the angle of elevation or depression
3. Identify the right-angled triangle formed by:
 - The vertical height difference
 - The horizontal distance
 - The line of sight (hypotenuse)
4. Use SOH CAH TOA to find the unknown

Common Formulas

For an object above the observer (elevation):

- $\tan(\theta) = \text{height} / \text{horizontal distance}$
- $\text{height} = \text{horizontal distance} \times \tan(\theta)$
- $\text{horizontal distance} = \text{height} / \tan(\theta)$

For an object below the observer (depression):

- $\tan(\theta) = \text{height} / \text{horizontal distance}$ (same formula!)
- The height is the difference in vertical elevation between observer and object

Worked Example

Problem: A building is 50 metres tall. From the top of the building, the angle of depression of a car on the ground is 25 degrees. How far is the car from the base of the building?

Step 1: Draw and label a diagram.

- Building height (vertical) = 50 m
- Angle of depression from top = 25 degrees
- Distance from base (horizontal) = ? (call it 'd')

The angle of depression from the top = angle of elevation from the bottom (alternate angles).
So the angle inside the triangle at the top (from horizontal down to car) = 25 degrees.

Alternatively, draw the triangle with the right angle at the base of the building:

- Opposite (height) = 50 m
- Adjacent (horizontal distance) = d
- Angle at the top (between vertical and hypotenuse) is $(90 - 25) = 65$ degrees
- Or simpler: the angle at the ground between horizontal and line of sight = 25 degrees (alternate angles)

Step 2: Identify the trig ratio.

We know the height opposite the 25-degree angle? Let's clarify:

- From the car looking up: angle of elevation = 25 degrees
- Opposite to this angle = building height = 50 m
- Adjacent to this angle = horizontal distance = d
- So $\tan(25 \text{ degrees}) = 50 / d$

Step 3: Substitute and solve.

$\tan(25 \text{ degrees}) = 0.4663$ (from calculator)

$$0.4663 = 50 / d$$

$$d = 50 / 0.4663$$

$$d = 107.2 \text{ m}$$

Answer: The car is approximately 107.2 metres from the base of the building.

Check: $50 / \tan(25) = 50 / 0.4663 = 107.2$. Reasonable for a 50 m building.

Common Mistakes

X Confusing which angle to use in the triangle — putting the angle of elevation/depression in the v

OK Fix: Draw a clear horizontal line from the observer's eye. The angle of elevation is above this line; the angle of depression is below this line.

X Forgetting that the observer's eye height may need to be added if the problem mentions it

OK Fix: If the problem says 'from eye level' or gives the observer's height, add that to the calculated height. Remember to label the diagram.

X Using the wrong trigonometric ratio (e.g., using sin instead of tan)

OK Fix: In elevation/depression problems, you are typically working with the vertical height (opposite) and horizontal distance (adjacent), so use $\tan$.

Summary

- Angle of elevation is measured upward from the horizontal; angle of depression is measured downward from the horizontal.
- The angle of depression from one point equals the angle of elevation from the other point (alternate angles are equal).
- Draw a diagram first — mark the horizontal line, the line of sight, the vertical height, and the horizontal distance.
- Most problems use $\tan(\theta)$ since the vertical height and horizontal distance form the opposite and adjacent sides.
- Always check whether the question asks for height, horizontal distance, or line-of-sight distance, and round appropriately.

E1_Q4: Bearing Problems

Hook

How does a ship's captain navigate safely through a busy shipping lane, or how does an air traffic controller direct planes to avoid collisions? The answer lies in **bearings** — a precise way of describing direction using angles measured clockwise from North. Bearings turn trigonometry into a practical tool for navigation, search-and-rescue, and even orienteering. One wrong degree can put a ship kilometres off course, so accuracy matters.

Prerequisites

- Understanding of angles measured clockwise from North (0 to 360 degrees)
- Basic trigonometry (SOH CAH TOA) for finding sides and angles in right-angled triangles
- Sine Rule and Cosine Rule for non-right-angled triangles
- Interpreting 3-figure bearings (e.g., 047 degrees, 180 degrees, 315 degrees)

Core Explanation

What Are Bearings?

A **bearing** is a direction given as an angle measured **clockwise from North**. It is always written as **three figures** (e.g., 047 degrees, not 47 degrees).

Rules for Bearings:

1. Measured from **North** (0 degrees)
2. Measured **clockwise**
3. Always written as **three digits** (e.g., 007 degrees, 045 degrees, 120 degrees, 270 degrees)

Key Bearing Facts:

- North = 000 degrees or 360 degrees
- East = 090 degrees
- South = 180 degrees
- West = 270 degrees

Back Bearings

The **back bearing** (or reverse bearing) is the bearing from B to A, given the bearing from A to B.

- If the bearing from A to B is less than 180 degrees: back bearing = bearing + 180 degrees
- If the bearing from A to B is 180 degrees or more: back bearing = bearing - 180 degrees

Solving Bearing Problems with Trigonometry

Bearing problems often involve:

- Drawing a diagram with North lines at each point
- Finding distances between points (using Pythagoras, Sine Rule, or Cosine Rule)
- Finding bearings between points (using inverse trig functions)

Common Triangle Setups

When a problem gives two bearings and a distance, you can:

1. Draw North lines at each relevant point
2. Use the fact that North lines are parallel to find angles inside the triangle
3. Apply the appropriate trigonometric method

North lines are always parallel — use this to find angles within the triangle by applying co-interior/alternate angle rules.

Worked Example

Problem: A ship sails from port A on a bearing of 060 degrees for 30 km to point B. It then changes course and sails on a bearing of 150 degrees for 40 km to point C. Find the distance from A to C and the bearing of C from A.

Step 1: Draw a diagram.

- From A: North line, then bearing 060 degrees, go 30 km to B
- From B: North line (parallel to North at A), then bearing 150 degrees, go 40 km to C
- Need AC and bearing of C from A

Step 2: Find angle ABC inside the triangle.

- Interior angle at B: The North line at B is parallel to the North line at A
- The bearing from A to B is 060 degrees, so angle between North and AB is 60 degrees

- At B, the bearing to C is 150 degrees, but AB points FROM B to A (reverse of 060)
- Back bearing B to A = $060 + 180 = 240$ degrees
- Difference between 240 and 150 = 90 degrees... let's think more carefully.

Let's use a simpler approach:

- At B, the angle between the North line and BA (pointing back to A) is the back bearing of AB
- AB bearing from A to B = 060, so BA bearing from B to A = $060 + 180 = 240$ degrees
- Bearing from B to C = 150 degrees
- Angle ABC = difference between BC bearing and BA bearing
- Going from BA (240) clockwise to BC (150): $(360 - 240) + 150 = 270$... that doesn't seem right either.

Let me recalculate carefully:

- North at B points up. AB goes from B to A at bearing $060 + 180 = 240$ degrees.
From North (000) clockwise to BA = 240 degrees. So the angle from North to BA is 240 degrees clockwise.
- But the interior angle of the triangle at B is the angle between BA and BC.
From BA (direction 240, or 120 degrees clockwise from South) to BC (direction 150).
The angle from North clockwise to BC = 150 degrees.
- The difference between 240 and 150 going the smaller way: $|240 - 150| = 90$ degrees.
- So angle ABC = 90 degrees (the triangle is right-angled at B!)

****Step 3:**** Find AC using Pythagoras.

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 30^2 + 40^2 = 900 + 1600 = 2500$$

$$AC = \sqrt{2500} = 50 \text{ km}$$

****Step 4:**** Find angle BAC.

$$\tan(\text{BAC}) = \text{opposite/adjacent} = BC/AB = 40/30 = 4/3$$

$$\text{angle BAC} = \tan^{-1}(4/3) = 53.1 \text{ degrees (to 1 d.p.)}$$

****Step 5:**** Find bearing of C from A.

$$\text{Bearing from A to B} = 060 \text{ degrees}$$

$$\text{angle BAC} = 53.1 \text{ degrees (the angle from AB to AC inside the triangle)}$$

$$\text{Bearing of C from A} = \text{bearing of B from A} + \text{angle BAC} = 060 + 53.1 = 113.1 \text{ degrees}$$

****Answer:**** AC = 50 km, and the bearing of C from A is 113 degrees (to the nearest degree) or 113.1 degrees.

****Check:**** The right angle at B (difference between 240 and 150 bearings = 90 degrees) confirms our approach. 30-40-50 is a Pythagorean triple (3-4-5 x 10).

Common Mistakes

X Writing a bearing as a two-digit number (e.g., 47 degrees instead of 047 degrees)

OK Fix: Bearings must always have three figures. If the angle is less than 100 degrees, add a leading zero. 47 degrees becomes 047 degrees.

X Confusing the direction — measuring anti-clockwise or from a point other than North

OK Fix: Always measure bearings clockwise from North. A common trick: think of a compass — North is 000, East is 090, South is 180, West is 270.

X Forgetting that North lines at different points are parallel, leading to incorrect angle calculation

OK Fix: Always draw a North line at every point in the diagram. The North lines are parallel, so you can use alternate angles.

Summary

- A bearing is an angle measured clockwise from North, always given as three figures (e.g., 047 degrees, 120 degrees, 315 degrees).
- The back bearing (reverse direction) is found by adding 180 degrees if under 180, or subtracting 180 degrees if 180 or more.
- Always draw North lines at every point — they are parallel and help find the interior angles of the navigation triangle.
- Use Pythagoras for right-angled navigation triangles, and Sine Rule or Cosine Rule for non-right triangles.
- When giving a final bearing, state it as a three-figure angle and round appropriately as specified in the question.

E1_Q5: Sine Rule

Hook

What happens when you need to find a missing side or angle in a triangle that doesn't have a right angle? The Sine Rule is one of two powerful tools (along with the Cosine Rule) that extend trigonometry beyond right-angled triangles. Surveyors use it to map out irregular plots of land, and astronomers use it to calculate distances to stars by triangulation. Any triangle — no right angle required — can be solved using this rule.

Prerequisites

- Understanding of labelling triangles with A, B, C for angles and a, b, c for opposite sides
- Right-angled trigonometry (SOH CAH TOA) as a foundation
- Solving algebraic equations involving fractions and cross-multiplication
- Using a scientific calculator for sin values and inverse sine ($\sin^{-1}$)

Core Explanation

The Sine Rule

The Sine Rule relates the sides of any triangle to the sines of their opposite angles. It works for **any** triangle (right-angled, acute, or obtuse).

Formula

For a triangle with sides a, b, c and opposite angles A, B, C:

$$a/\sin A = b/\sin B = c/\sin C$$

Or equivalently:

$$\sin A/a = \sin B/b = \sin C/c$$

Where:

- **Side a** is opposite **angle A**
- **Side b** is opposite **angle B**
- **Side c** is opposite **angle C**

When to Use the Sine Rule

Use the Sine Rule when you know:

Case 1: Two angles and any side (ASA or AAS)

- Find the third angle (angles sum to 180 degrees)
- Use the Sine Rule to find the missing sides

Case 2: Two sides and a non-included angle (SSA or 'the ambiguous case')

- Use the Sine Rule to find an angle opposite a known side
- Be careful: there may be 0, 1, or 2 possible triangles (the ambiguous case)

The Ambiguous Case (SSA)

When given two sides and an angle **not between** them, check:

- If the angle is acute and the opposite side is shorter than the adjacent side, there may be **two possible triangles** (one acute, one obtuse)
- Remember: $\sin(\theta) = \sin(180^\circ - \theta)$, so two different angles can have the same sine value
- If the calculated angle is acute, also check whether $(180 - \text{that angle})$ is a valid triangle

Worked Example

Problem: In triangle ABC, angle A = 40 degrees, angle B = 60 degrees, and side a = 8 cm. Find side b.

Step 1: Identify what we know.

- A = 40 degrees, B = 60 degrees
- a = 8 cm (side opposite angle A)
- b = ? (side opposite angle B)

Step 2: Check we have enough information.

We have two angles and a side opposite one of them (AAS). Perfect for the Sine Rule.

We don't need angle C since we're using A and B.

Step 3: Write the Sine Rule for the relevant parts.

$$a/\sin A = b/\sin B$$

****Step 4:**** Substitute known values.

$$8/\sin(40 \text{ degrees}) = b/\sin(60 \text{ degrees})$$

****Step 5:**** Calculate the sine values.

$$\sin(40 \text{ degrees}) = 0.6428$$

$$\sin(60 \text{ degrees}) = 0.8660$$

****Step 6:**** Solve for b.

$$8 / 0.6428 = b / 0.8660$$

$$12.445 = b / 0.8660$$

$$b = 12.445 \times 0.8660$$

$$b = 10.78 \text{ cm}$$

****Answer:**** Side b is approximately 10.8 cm (to 3 significant figures).

****Step 7:**** Verify.

$$\text{Angle C} = 180 - 40 - 60 = 80 \text{ degrees}$$

$$\text{Check using the ratio: } a/\sin A = 8/0.6428 = 12.44$$

$$b/\sin B = 10.78/0.8660 = 12.44$$

$$c/\sin C = c/\sin 80 = c/0.9848 \text{ — should also equal } 12.44$$

$$\text{So } c = 12.44 \times 0.9848 = 12.25 \text{ cm. Consistent.}$$

Common Mistakes

X Using the Sine Rule in the wrong situation (e.g., when you have two sides and the included angle)

OK Fix: The Sine Rule needs matching angle-side pairs. If you know two sides and the angle BETWEEN them, you can use the Sine Rule.

X Forgetting the ambiguous case and only finding one solution when two exist

OK Fix: When using the Sine Rule to find an angle from two sides, check if both the acute angle and its supplement are possible.

X Mixing up which side pairs with which angle (e.g., using $a/\sin B$ instead of $a/\sin A$)

OK Fix: Remember: side a is opposite angle A, side b is opposite angle B. Label your triangle clearly before starting.

Summary

- The Sine Rule ($a/\sin A = b/\sin B = c/\sin C$) works for ANY triangle, not just right-angled ones.
- Use the Sine Rule when you know two angles and any side (ASA/AAS), or two sides and a non-included angle (SSA).
- Label your triangle carefully — side a is opposite angle A, and this pairing is essential for the formula.
- Beware of the ambiguous case (SSA): there may be 0, 1, or 2 possible triangles. Check both acute and obtuse possibilities.
- Always verify your answer by checking that the three angles sum to 180 degrees and that all ratios are equal.

E1_Q6: Cosine Rule

Hook

Imagine you're a pilot who knows the distance flown on two legs of a journey and the angle between them, but you need to know the direct distance back to your starting point. Or you know all three sides of a triangular field and need to find its corner angles. The Cosine Rule is the tool you need — it's like Pythagoras' Theorem for any triangle, with a clever adjustment to account for angles that aren't 90 degrees. It's essential for navigation, surveying, and engineering.

Prerequisites

- Pythagoras' Theorem ($a^2 + b^2 = c^2$) for right-angled triangles
- Labelling triangles with sides a, b, c opposite angles A, B, C
- Using a calculator for cosine values and inverse cosine ($\cos^{-1}$)
- Basic algebraic manipulation including square roots and rearranging formulas

Core Explanation

The Cosine Rule

The Cosine Rule is a generalisation of Pythagoras' Theorem that works for **any** triangle. When the included angle is 90 degrees, the Cosine Rule simplifies to Pythagoras' Theorem.

Formulas

Finding a side (given two sides and the included angle):

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Where:

- **a** is the side opposite angle A (the side you want to find)
- **b** and **c** are the other two sides
- **A** is the angle between sides b and c (the included angle)

Finding an angle (given three sides):

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

When to Use the Cosine Rule

Case 1: Two sides and the included angle (SAS) — use the first formula to find the third side.

- You know sides b and c , and the angle A between them
- Find side a using $a^2 = b^2 + c^2 - 2bc \cos A$

Case 2: Three sides (SSS) — use the second formula to find any angle.

- You know sides a, b, c
- Find angle A using $\cos A = (b^2 + c^2 - a^2) / (2bc)$

Connection to Pythagoras

If $A = 90$ degrees, then $\cos(90) = 0$, so:

$$a^2 = b^2 + c^2 - 2bc(0) = b^2 + c^2$$

This is just Pythagoras' Theorem. So the Cosine Rule is a generalisation.

Worked Example

****Problem:**** In triangle ABC, side $b = 7$ cm, side $c = 9$ cm, and the included angle $A = 50$ degrees. Find side a .

****Step 1:**** Identify what we know.

- $b = 7$ cm (side opposite angle B)
- $c = 9$ cm (side opposite angle C)
- $A = 50$ degrees (angle between sides b and c)
- $a = ?$ (side opposite angle A)

This is SAS (two sides and the included angle). Use the Cosine Rule.

****Step 2:**** Write the Cosine Rule formula.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

****Step 3:**** Substitute the known values.

$$a^2 = 7^2 + 9^2 - 2(7)(9) \cos(50 \text{ degrees})$$

****Step 4:**** Calculate each part.

$$7^2 = 49$$

$$9^2 = 81$$

$$2(7)(9) = 126$$

$$\cos(50 \text{ degrees}) = 0.6428$$

$$a^2 = 49 + 81 - 126(0.6428)$$

$$a^2 = 130 - 81.00$$

$$a^2 = 49.00$$

****Step 5:**** Take the square root.

$$a = \sqrt{49.00}$$

$$a = 7.00 \text{ cm}$$

****Answer:**** Side a is 7.00 cm.

****Check:**** $b = 7$, $a = 7$, $c = 9$. The triangle is isosceles with $a = b$. This makes sense because $\cos(50) = 0.6428$ and... wait, let me recheck:

$$a^2 = 49 + 81 - 126 \times 0.6428$$

$$a^2 = 130 - 81.00$$

$$a^2 = 49.00 \text{ — yes, } a = 7.00 \text{ cm.}$$

Let's verify using the Sine Rule:

$$a/\sin A = 7/\sin(50) = 7/0.7660 = 9.14$$

$$b/\sin B = 7/\sin B = 9.14 \text{ so } \sin B = 7/9.14 = 0.7660, B = 50 \text{ degrees.}$$

So $A = B = 50$ degrees and $C = 80$ degrees.

$$c/\sin C = 9/\sin(80) = 9/0.9848 = 9.14. \text{ Consistent.}$$

Common Mistakes

X Using the Cosine Rule when the angle is NOT the included angle between the two known sides

OK Fix: The Cosine Rule requires the angle BETWEEN the two sides you know (SAS). If the angle is opposite one

X Forgetting to take the square root after finding a^2 , leaving the answer as a^2

OK Fix: The formula gives a^2 , not a . Always take the positive square root to find the side length. Write $a = \sqrt{\dots}$

X Getting the formula wrong — particularly the minus sign or the factor of 2

OK Fix: Memorise: $a^2 = b^2 + c^2 - 2bc \cos A$. The minus sign is crucial. For the angle formula, remember: $\cos A = (b^2 + c^2 - a^2) / (2bc)$

Summary

- The Cosine Rule ($a^2 = b^2 + c^2 - 2bc \cos A$) generalises Pythagoras to any triangle and reduces to it when $A = 90$ degrees.
- Use the Cosine Rule for SAS (two sides and the included angle) to find the third side.
- Use the rearranged form $\cos A = (b^2 + c^2 - a^2) / (2bc)$ for SSS (three sides) to find an angle.
- Remember to take the square root when finding a side, and use inverse cosine ($\cos^{-1}$) when finding an angle.
- Label your triangle carefully: side a is opposite angle A , and A must be the included angle between sides b and c .

E1_Q7: Area of Triangle $\frac{1}{2}ab \sin C$

Hook

You know that area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$ — but what if you can't measure the height? What if you're a land surveyor who knows two sides of a triangular plot and the angle between them, but can't walk across the property to measure the perpendicular height? The formula $\text{Area} = \frac{1}{2} ab \sin C$ solves this problem elegantly. It's also how engineers calculate the area of irregular surfaces and why GPS can estimate the area of any triangle on Earth's surface without needing altitude data.

Prerequisites

- Basic formula for area of a triangle ($\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$)
- Sine of an angle (SOH CAH TOA foundation)

- Labelling triangles with sides a , b , c opposite angles A , B , C
- Using a calculator for sine values

Core Explanation

Area of a Triangle Using Trigonometry

The standard formula $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ requires you to know the perpendicular height. The trigonometric formula removes this requirement by using two sides and the included angle.

Formula

For any triangle ABC:

$$\text{Area} = \frac{1}{2} ab \sin C$$

Where:

- a and b are two sides of the triangle
- C is the **included angle** (the angle between sides a and b)

Why Does This Work?

In triangle ABC, if we take side b as the base, the perpendicular height from vertex A to side b is:

- height = $c \times \sin B$ (using the right triangle formed by dropping a perpendicular)
- Or equivalently, height = $a \times \sin C$

Substituting $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$:

$$\text{Area} = \frac{1}{2} \times b \times (a \times \sin C) = \frac{1}{2} ab \sin C$$

Three Equivalent Forms

You can use any version depending on which angle and sides you know:

- $\text{Area} = \frac{1}{2} ab \sin C$ (using sides a , b and angle C between them)
- $\text{Area} = \frac{1}{2} bc \sin A$ (using sides b , c and angle A between them)
- $\text{Area} = \frac{1}{2} ac \sin B$ (using sides a , c and angle B between them)

Key Reminder

The angle must be the **included angle** — the angle between the two sides you are using. Side a and side b meet at angle C , so they 'include' angle C .

Worked Example

Problem: A triangular field has sides of length 50 m and 60 m, with an included angle of 40 degrees. Find the area of the field.

Step 1: Identify what we know.

- Side a = 50 m (one side)
- Side b = 60 m (the other side)
- Angle C = 40 degrees (the angle between sides a and b)

Step 2: Write the formula.

$$\text{Area} = \frac{1}{2} ab \sin C$$

Step 3: Substitute the known values.

$$\text{Area} = \frac{1}{2} \times 50 \times 60 \times \sin(40 \text{ degrees})$$

Step 4: Calculate step by step.

$$\frac{1}{2} \times 50 \times 60 = \frac{1}{2} \times 3000 = 1500$$

$$\sin(40 \text{ degrees}) = 0.6428$$

$$\text{Area} = 1500 \times 0.6428$$

$$\text{Area} = 964.2 \text{ m}^2$$

Answer: The area of the field is approximately 964 square metres (to 3 significant figures).

Check using alternative form:

Let's verify by finding the third side first (using Cosine Rule) and then using Heron's formula.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 2500 + 3600 - 2(50)(60)(0.7660)$$

$$c^2 = 6100 - 4596 = 1504$$

$$c = 38.79 \text{ m}$$

$$\text{Heron's formula: } s = (50 + 60 + 38.79)/2 = 74.395$$

$$\text{Area} = \sqrt{74.395 \times 24.395 \times 14.395 \times 35.605}$$

$$\text{Area} = \sqrt{74.395 \times 24.395 \times 14.395 \times 35.605}$$

Let's just check: 964.2 seems reasonable for a triangle with sides 50 and 60 at 40 degrees.

Common Mistakes

X Using an angle that is NOT the included angle between the two given sides

OK Fix: The formula requires the angle BETWEEN the two sides. For example, if you know sides a and b, you must use the angle between them.

X Forgetting to multiply by 1/2 (or dividing by 2 instead of multiplying by 0.5)

OK Fix: $\text{Area} = \frac{1}{2} ab \sin C$. The 1/2 is essential — don't skip it! One way to remember: the standard area formula for a rectangle is $\text{Area} = \text{base} \times \text{height}$, and a triangle is half of a rectangle.

X Using degrees in the formula but having the calculator in radian mode, giving a wildly wrong answer

OK Fix: Always check your calculator is in DEGREE mode. In radian mode, $\sin(40) = 0.745$ (for 40 radians!) instead of 0.6428.

Summary

- $\text{Area} = \frac{1}{2} ab \sin C$ calculates the area of any triangle given two sides and the included angle (SAS).
- The angle must be the included angle — the one between the two known sides.

- There are three equivalent forms: $\frac{1}{2} ab \sin C$, $\frac{1}{2} bc \sin A$, and $\frac{1}{2} ac \sin B$ — pick the one that matches your known sides and angle.
- The formula is derived from the standard $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ by expressing the height in terms of a side and the sine of an angle.
- Always include square units (e.g., m^2 , cm^2) in your answer and ensure your calculator is in degree mode.